

Navier–Stokes Existence and Smoothness: Two Computational Probes with PINNeAPple

Physics-informed search for self-similar blow-up on a model with a known answer, and a resolved Taylor–Green vortex

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Abstract

The Clay problem asks whether smooth, finite-energy solutions of the 3-D incompressible Navier–Stokes equations stay smooth for all time. No computation can settle it; we test two tools that are used in the search for candidate singularities. (A) On inviscid Burgers, where smooth self-similar blow-up profiles exist only for $\lambda = 1/(2i + 2)$, a PINNeAPple PINN with a smoothness penalty did not identify the admissible rates: across $\lambda \in [0.2, 0.7]$ the residual grew monotonically with λ (factor 8.3) and had no minimum at 0.25 or 0.5, and with λ trainable the network converged to $\lambda = 0.418$ and 0.019 instead of 0.5 and 0.25. This is a negative result: the penalty was too weak to exclude non-smooth solutions, which exist for every λ . (B) A pseudo-spectral DNS of the Taylor–Green vortex at $\text{Re} = 1600$ on $48^3, 64^3, 96^3$ grids keeps the energy balance $-dE/dt = \varepsilon$ to a relative error of at most $2.8e-04$; the peak dissipation on the finest grid is 0.0134 at $t = 8.82$, versus about 0.0127 near $t \approx 9$ in published high-resolution DNS. Vorticity stays bounded over the run, as expected at finite Re .

Keywords: Navier–Stokes; blow-up; self-similar solutions; PINN; Taylor–Green vortex; DNS; Millennium Prize problems

What this study shows — and what it does not

- Does not prove or disprove global regularity; nothing computed at finite resolution and finite time can.
- Part A is a validation of a method on a model with a known answer, and the method failed that validation as configured. It should not be used on Navier–Stokes (or Euler) without a stronger admissibility criterion.
- Part B verifies the solver (energy balance, grid dependence) on a flow that is known to stay smooth; it is a baseline, not a blow-up search.

1 Background

For $u_t + u u_x = 0$ the self-similar ansatz $u = (T - t)^\lambda U(\xi)$, $\xi = x/(T - t)^{1+\lambda}$ gives $-\lambda U + (1 + \lambda)\xi U' + UU' = 0$ with $U(0) = 0$, $U'(0) = -1$. Solutions exist for every $\lambda > 0$, but they are smooth (analytic) only for $\lambda = 1/(2i + 2)$; for $\lambda = 1/2$ the profile is the real root of $\xi = -U - U^3$ [1, 2]. This is the setting in which physics-informed networks were recently used to find unstable self-similar solutions of fluid equations [3], so it is the natural first test.

$$-\lambda U + (1 + \lambda) \xi U' + U U' = 0, U(0) = 0, U'(0) = -1 \quad (1)$$

2 Methods

- (A) PINNeAPple ModifiedMLP (4×64 , 16 Fourier features); $U = -\xi + \xi s^2 g(s^2)$ with $s = \xi/8$, so oddness and $U'(0) = -1$ hold exactly. Loss: mean squared residual + $10^{-6} \cdot \text{mean} (U''')^2$ as the smoothness penalty; 6000 Adam steps then L-BFGS; residual measured on 4001 fresh points. λ fixed on a grid, then λ trainable from 0.45 and from 0.22.
- (B) PINNeAPple SpectralNS3D: periodic box $[0, 2\pi]^3$, rotational form, 2/3-rule dealiasing, RK4 with an exact viscous integrating factor; $u = (\sin x \cos y \cos z, -\cos x \sin y \cos z, 0)$, $\nu = 1/1600$, $t \in [0, 10]$. Diagnostics: E , enstrophy Z , $\varepsilon = 2\nu Z$, max $|u|$ and the Beale–Kato–Majda integral $\int \max |u| dt$ [4].

3 Results

3.1 A. Self-similar Burgers profiles

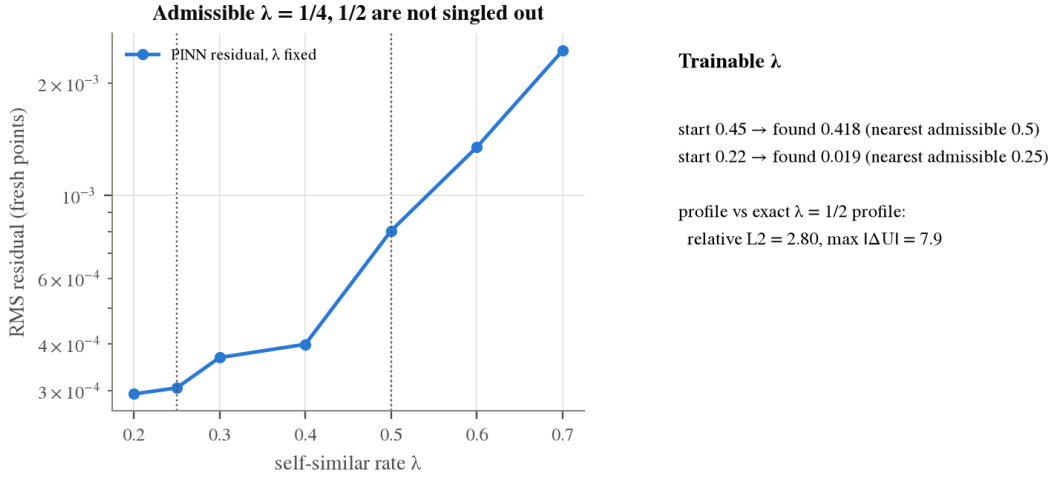


Figure 1. (A) Left: residual with λ fixed; dotted lines mark the admissible rates $1/4$ and $1/2$. Right: trainable- λ outcomes.

Table 1. Residual with λ fixed. A useful method would show clear minima at 0.25 and 0.5 .

λ	RMS residual (fresh points)
0.2	2.94e-04
0.25	3.05e-04
0.3	3.68e-04
0.4	4.00e-04
0.5	8.05e-04
0.6	1.35e-03
0.7	2.45e-03

The residual is small for every λ (at most $2.4\text{e-}03$): the network fits a solution of the ODE for any rate, as it can, since non-smooth solutions exist for all λ . The residual rises steadily with λ and has no dip at the admissible rates, so the smoothness penalty did not separate them. With λ trainable the rate drifted to 0.418 (from 0.45) and 0.019 (from 0.22). The profile obtained from the first run differs from the exact $\lambda = 1/2$ profile by a relative L2 error of 2.80 , so it is a different solution, not an inaccurate version of the right one.

3.2 B. Taylor–Green vortex at $\text{Re} = 1600$

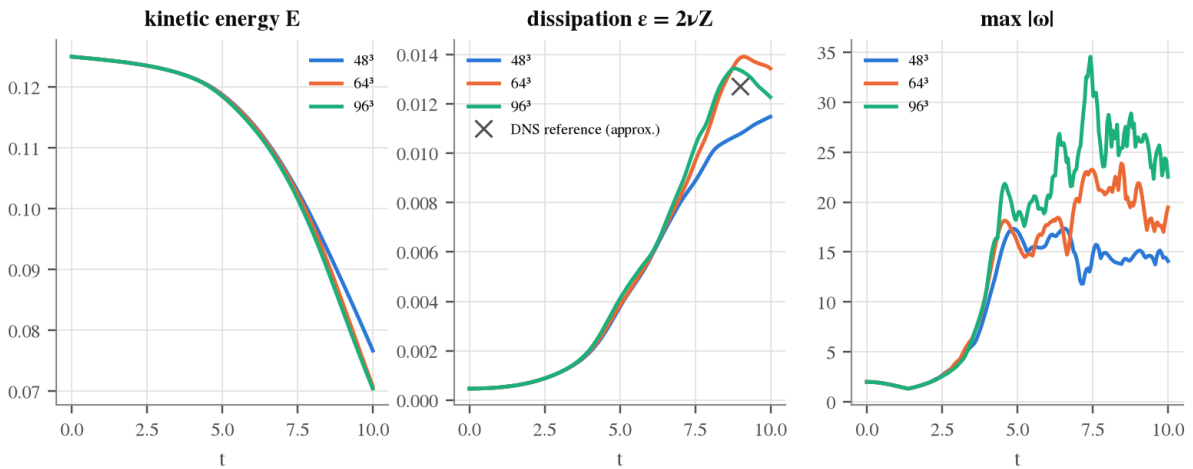


Figure 2. (B) Energy, dissipation and maximum vorticity on each grid; $\times$ marks the approximate peak of published high-resolution DNS.

Table 2. Grid dependence. "(end)": the maximum occurred at the last time step, so the dissipation was still rising.

grid	wall time (s)	peak ϵ	t at peak ϵ	energy-balance rel. error	max $ \omega $
48^3	140	0.0115	10.00 (end)	1.9e-04	17.4
64^3	1244	0.0139	9.10	2.3e-04	23.9
96^3	40305	0.0134	8.82	2.8e-04	34.6

The energy balance holds to within $2.8\text{e-}04$ on every grid, so the time integration and dissipation diagnostic are consistent. Relative to the reference peak, the computed peak dissipation deviates by -10% (48^3), $+10\%$ (64^3), $+6\%$ (96^3); the deviation does not shrink monotonically: the coarsest grid under-predicts the peak and a finer one over-predicts it, the usual non-monotone behaviour of an under-resolved spectral simulation near the dissipation peak. Grids of 48^3 – 96^3 under-resolve this flow near its dissipation peak (published DNS uses 256^3 – 512^3), so these numbers are a resolution study, not a converged answer. The maximum vorticity also grows with resolution (17, 24, 35), so it is not converged either; finer grids resolve thinner vortex sheets. It stays finite on every grid, which at $\text{Re} = 1600$ is expected and says nothing about the inviscid or high-Re limit.

4 Discussion

The self-similar probe is the more relevant tool for the Millennium question, and it is the one that failed here. What it lacks is an admissibility criterion strong enough to exclude the non-smooth solution family, for example imposing the analytic structure of U (expansion in ξ with the exact λ -dependent exponents) or demanding a smooth continuation beyond the finite ξ -window, as in [3]. We report the failure because a PINN that appears to converge for every λ would give false confidence if applied directly to Euler or Navier–Stokes. The DNS part is a verified baseline for later blow-up candidates, not evidence about regularity.

References

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